Drug review classification

Vanilla LSTM, CIFG and Working Memory Connections

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Advanced NLP, 2026/2027

Task and labels

Input: review text only

Ratings 1-4: Negative

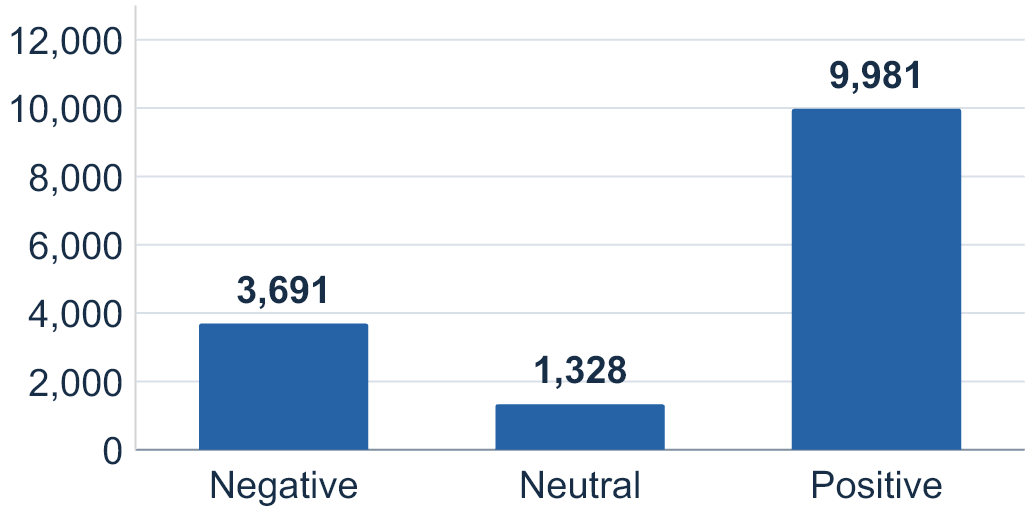
Ratings 5-6: Neutral

Ratings 7-10: Positive

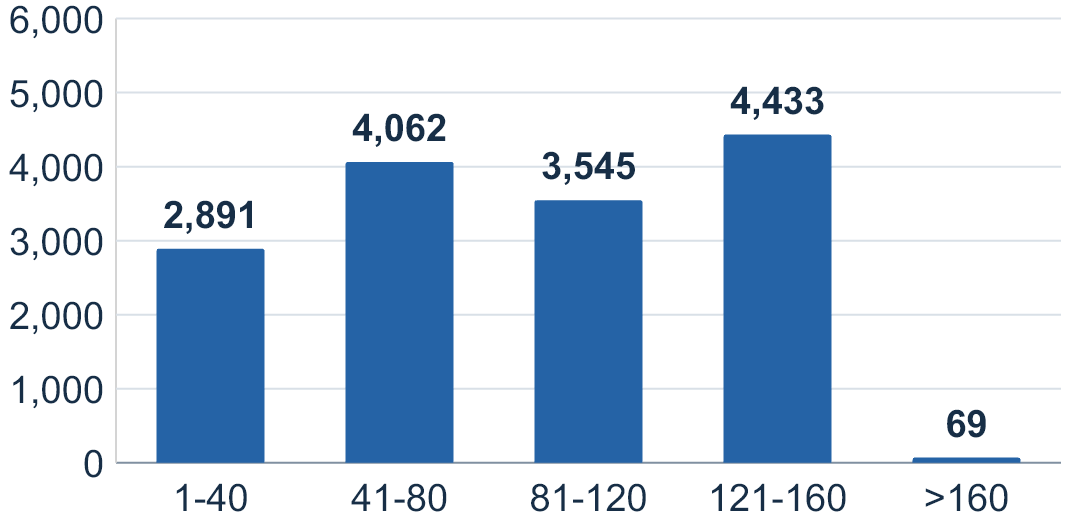
Ratings define targets and never enter the model

Class imbalance and review lengths

Training class counts



Review length in tokens



66.54% Positive and 8.85% Neutral in training

160 tokens retain complete text for 99.54% of training reviews

Duplicate leakage and shared partitions

32,111 cleaned training texts also appear in raw test

Exclude these texts before splitting training data

15,000 training, 3,000 validation, 48,257 test reviews

Same partitions and train-only vocabulary for all models

Three custom recurrent cells

Vanilla: independent input, forget and output gates

CIFG: input gate i = 1 - f

WMC: tanh memory projections enter the gates

WMC output gate reads the updated memory c_t

All linear operations use nn.Linear

Parameter cost

Cell	Recurrent parameters	Total parameters
Vanilla	12,416	780,515
CIFG	9,312	777,411
WMC	15,488	783,587

CIFG reduces recurrent parameters by 25%

The shared embedding has 768,000 parameters

Hyperparameter study

Setting	Vanilla val F1	CIFG val F1	WMC val F1
fixed	0.5425	0.5558	0.5571
lr_0.001	0.5576	0.5652	0.5447
batch_128	0.5657	0.5547	0.5556
adam	0.3389	0.2663	0.2856

Fixed: AdamW, LR 0.003, batch 256

One-factor changes: LR 0.001, batch 128, Adam

Six-epoch maximum and validation-based early stopping

Held-out performance

Cell	Fixed accuracy	Fixed F1	Selected setting	Selected F1
Vanilla	65.80%	0.5501	batch_128	0.5583
CIFG	67.78%	0.5484	lr_0.001	0.5650
WMC	70.41%	0.5583	fixed	0.5583

All scores use the same 48,257 test reviews

Selected settings follow validation Macro-F1

Errors and limits

Vanilla selected

True	Neg	Neu	Pos
Neg	6551	3074	2468
Neu	898	1620	1816
Pos	1737	4772	25321

CIFG selected

True	Neg	Neu	Pos
Neg	8162	1893	2038
Neu	1464	1288	1582
Pos	3434	3825	24571

WMC selected

True	Neg	Neu	Pos
Neg	7488	2190	2415
Neu	1299	1263	1772
Pos	2706	3895	25229

Rows: true class. Columns: predicted class.

Neutral ratings cover mixed experiences

Single seed, reduced training data and compact hidden state

Exact deduplication leaves paraphrases and group risks

Conclusion and next experiments

Validation-selected candidate: Vanilla (vanilla_batch_128)

Custom cells support a controlled, measured comparison

Next: full training pool, repeated seeds and broader tuning

Pretrained embeddings and attention are useful extensions